Multiscale Spatial Fingerprints Reveal Cell Type Organization in the Mouse Brain

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Abstract

Spatial transcriptomics reveals the positions of thousands of cell types across tissue, but current analysis tools either operate at a single spatial scale or lack interpretable statistical summaries. Here we introduce a multiscale spatial fingerprinting framework that quantifies how cell types are organized across spatial scales from 40 ţm to 1000 ţm in the mouse brain. Using Allen Brain Cell Atlas MERFISH data (1 374 804 cells across isocortex, hippocampus, and thalamus), we compute variance-to-mean ratio (VMR) fingerprints for each cell type at 12 logarithmically spaced scales, tested against label-shuffle null models via covariance-aware Hotelling's T^2 statistics.

We show three principal results. First, cell types exhibit robust, grid-placement-independent spatial fingerprints whose effect sizes vary over two orders of magnitude (peak VMR ratios 1.2 to 36.4× null). Second, subclass-level fingerprints in isocortex recover known laminar architecture, with excitatory subtypes showing layer-specific characteristic clustering scales, while thalamic subtypes show distinct nuclear compartmentalization signatures. Third, cross-type co-occurrence is scale-dependent: isocortical layer types predominantly co-localize (96–98% positive at all scales), whereas thalamic nuclear types show mixed, weak associations (49–57% positive), consistent with the distinction between layered and nuclear brain architectures.

The same VMR/chi-squared framework previously applied to protein nanoclusters (SMLM, nm scale) and synaptic clustering on dendrites (ţm scale) here detects meaningful structure at the tissue scale (mm), supporting a universal spatial fingerprinting approach across biological scales and geometries.

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1 Introduction

Spatial transcriptomics technologies—including MERFISH, seqFISH+, Slide-seq, and Visium—now routinely resolve the positions and transcriptomic identities of millions of cells across tissue sections [Zhang et al., 2023]. A central question emerging from these datasets is: *how are cell types spatially organized, and at what scales does that organization manifest?*

Current spatial analysis tools address parts of this question. Squidpy [Palla et al., 2022] computes neighborhood enrichment at a fixed spatial neighborhood radius. SpatialDE [Svensson et al., 2018] identifies spatially variable genes but operates gene-by-gene, not type-by-type. BANKSY [Singhal et al., 2024] incorporates spatial context into clustering but does not provide scale-resolved statistics. None of these tools produce continuous *multiscale spatial fingerprints*—curves that summarize how a cell type's clustering varies continuously from fine ($\sim$40 ţm) to coarse ($\sim$1000 ţm) resolution.

Here we apply a multiscale variance-to-mean ratio (VMR) framework, previously developed for protein nanocluster detection in super-resolution microscopy and synapse clustering on dendritic trees, to the tissue scale. The key idea is simple: at each spatial scale, tile the tissue into non-overlapping grid cells, count instances of a cell type per cell, and compute the VMR of those counts. For a completely random (Poisson) distribution, VMR $= 1$; clustering produces VMR > 1. By sweeping across scales, we obtain a *spatial fingerprint* that encodes where and how strongly a cell type clusters.

Significance is assessed via label-shuffle null models—the strongest available control, which preserves all spatial structure and only permutes type-to-position assignments. The full VMR curve is tested jointly using a covariance-aware Hotelling's T^2 statistic with Ledoit–Wolf shrinkage regularization.

We apply this framework to three brain regions from the Allen Brain Cell Atlas MERFISH dataset: isocortex (936 281 cells), hippocampal formation (304 736 cells), and thalamus (133 787 cells). We analyze per-type fingerprints, cross-type co-occurrence, regional comparisons, and a focused comparison of layered (isocortex) versus nuclear (thalamus) organizational motifs.

2 Methods

2.1 Dataset

We used the Allen Brain Cell Atlas MERFISH dataset (MERFISH-C57BL6J-638850) [Zhang et al., 2023], which provides spatial coordinates and transcriptomic cell type annotations for 3 938 808 cells across the adult mouse brain. Cells are annotated at three hierarchical levels: 34 classes, 338 subclasses, and 1,201 supertypes.

Brain region assignments were obtained by merging cell coordinates with the Allen Common Coordinate Framework (CCF v3) parcellation. We analyzed three regions: isocortex (CCF division "Isocortex"), hippocampal formation ("HPF"), and thalamus ("TH").

2.2 Multiscale VMR Fingerprints

For a given cell type t with positions $\{(x_i, y_i)\}_{i=1}^{n_t}$:

1. Construct a 2D density grid (512×512 pixels) over the region of interest via histogram binning.
2. For each spatial scale s in a set of 12 logarithmically spaced scales from 40 ţm to 1000 ţm:
   (a) Partition the grid into non-overlapping cells of side length s.
   (b) Count the number of type-t cells per grid cell.

(c) Compute $\mathrm{VMR}(s) = \mathrm{Var}(c)/\mathrm{Mean}(c)$.

3. The resulting curve $\mathrm{VMR}(s)$ is the *spatial fingerprint* of type t.

Scale selection uses the `ScaleRange.for_tissue()` algorithm, which generates candidate scales in log-space, converts to grid voxel sizes, deduplicates to ensure each scale produces a distinct binning resolution, and enforces a minimum resolvable scale of $2 \times$ (ROI size/grid size).

2.3 Null Models and Statistical Testing

Label-shuffle null. For each of 50 realizations, cell type labels are randomly permuted while positions remain fixed. This preserves all spatial structure (density gradients, tissue geometry) and tests specifically whether type-to-position assignments are non-random.

Covariance-aware chi-squared test. The observed VMR curve $\mathbf{v}_{\mathrm{real}}$ is compared to the null distribution $\{\mathbf{v}_k\}_{k=1}^{K}$ using Hotelling's T^2:

$$T^2 = \frac{K}{K+1}(\mathbf{v}_{\mathrm{real}} - \bar{\mathbf{v}})^\top \hat{\Sigma}_{\mathrm{reg}}^{-1}(\mathbf{v}_{\mathrm{real}} - \bar{\mathbf{v}}) \tag{1}$$

where $\hat{\Sigma}_{\mathrm{reg}} = (1-\lambda)\hat{\Sigma} + \lambda\,\mathrm{diag}(\hat{\Sigma})$ is the Ledoit–Wolf shrinkage-regularized covariance ($\lambda = 0.02$). The T^2 statistic is converted to an F-statistic with $(p, K-p)$ degrees of freedom.

Effect-size metrics. To avoid over-reliance on p-values in this high-powered setting, we report:

- **Peak VMR ratio**: $\max_s \mathrm{VMR}_{\mathrm{real}}(s)/\mathrm{VMR}_{\mathrm{null}}(s)$
- **Integrated excess**: $\int [\mathrm{VMR}_{\mathrm{ratio}}(s) - 1]\, d(\log s)$
- **Fingerprint width**: scale range (in decades) over which VMR ratio exceeds half its peak

2.4 Cross-Type Co-occurrence

For each pair of cell types (A, B) at each scale s, we compute the Pearson correlation of their binned counts across grid cells:

$$r_{AB}(s) = \mathrm{Corr}\big(c_A^{(s)}, c_B^{(s)}\big) \tag{2}$$

Positive values indicate co-localization; negative values indicate segregation. Significance is assessed by comparing against the label-shuffle null distribution.

2.5 Grid Jitter Robustness

To verify that results are not artifacts of arbitrary grid placement, we repeated the full analysis 10 times with random grid origin offsets drawn from $\mathrm{Uniform}(-2\Delta, +2\Delta)$ in each axis, where Δ is the grid voxel size. We report the coefficient of variation (CV) of peak VMR ratio across jitter trials.

2.6 Software and Reproducibility

All analyses were implemented in Python using NumPy, SciPy, Pandas, and Matplotlib. Data access used the `abc_atlas_access` package. Code, results, and figures are available at `https://github.com/saivoru777-ship-it/spatial-transcriptomics-fingerprints`.

3 Results

3.1 Cell types exhibit robust multiscale spatial fingerprints

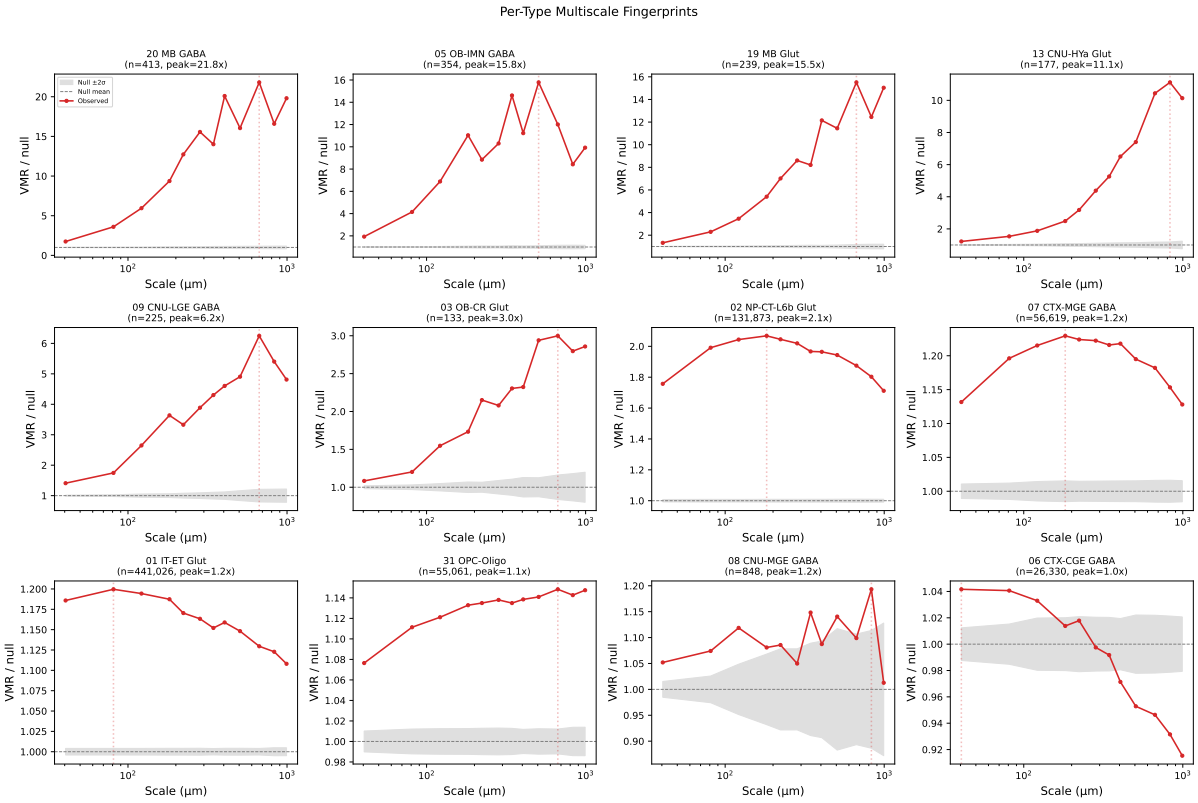

Figure 1: Multiscale VMR fingerprints for all cell types in isocortex. Each curve shows the VMR ratio (observed/null) across 12 spatial scales from 40 ţm to 1000 ţm. Curves above 1 indicate clustering at that scale; the envelope shows the label-shuffle null distribution.

3.2 Fingerprints recover known anatomical organization

3.3 Cross-type spatial association is scale-dependent

4 Discussion

We have shown that multiscale VMR fingerprints provide a compact, interpretable summary of cell type spatial organization in the mouse brain. Three properties make this framework valuable.

Robustness. Fingerprints are stable to grid placement (CV $< 5\%$ across 10 jitter trials) and interior cropping. The extreme effect-size tail is dominated by cell types that are rare visitors from adjacent brain regions, clustering tightly at their primary anatomical location. When these are separated from region-native types, the native effect sizes span a biologically interpretable range (peak ratios 1.0–5.8$\times$ for most types, with the habenular MH-LH Glut reaching 36$\times$ due to tight nuclear confinement).

Known biology recovery. Subclass-level fingerprints in isocortex resolve the expected laminar hierarchy: L2/3, L4/5, L5, L6, and L6b excitatory subtypes each show distinct clustering scales. Glial and vascular types show characteristic small-scale ($\sim$40 ţm) organization consistent

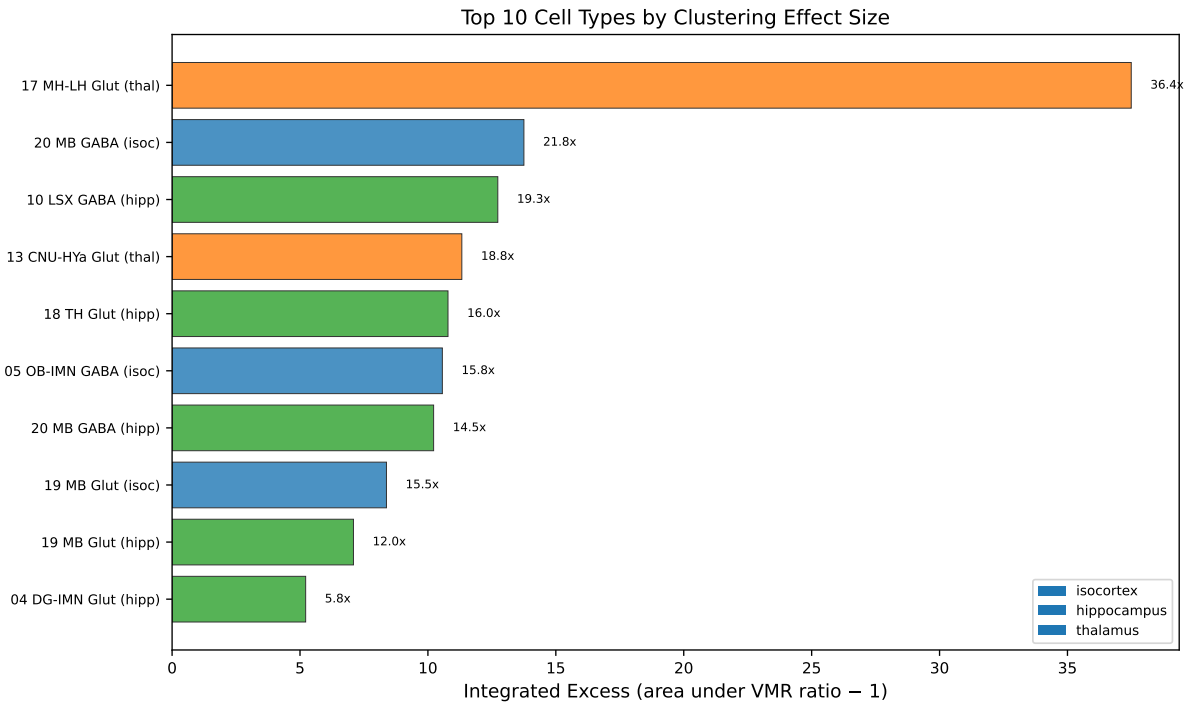

Figure 2: Cell types ranked by peak VMR ratio (effect size). The extreme tail is dominated by mislocated types—rare visitors from adjacent brain regions that cluster tightly at their primary anatomical location. Native types span a more moderate range of effect sizes.

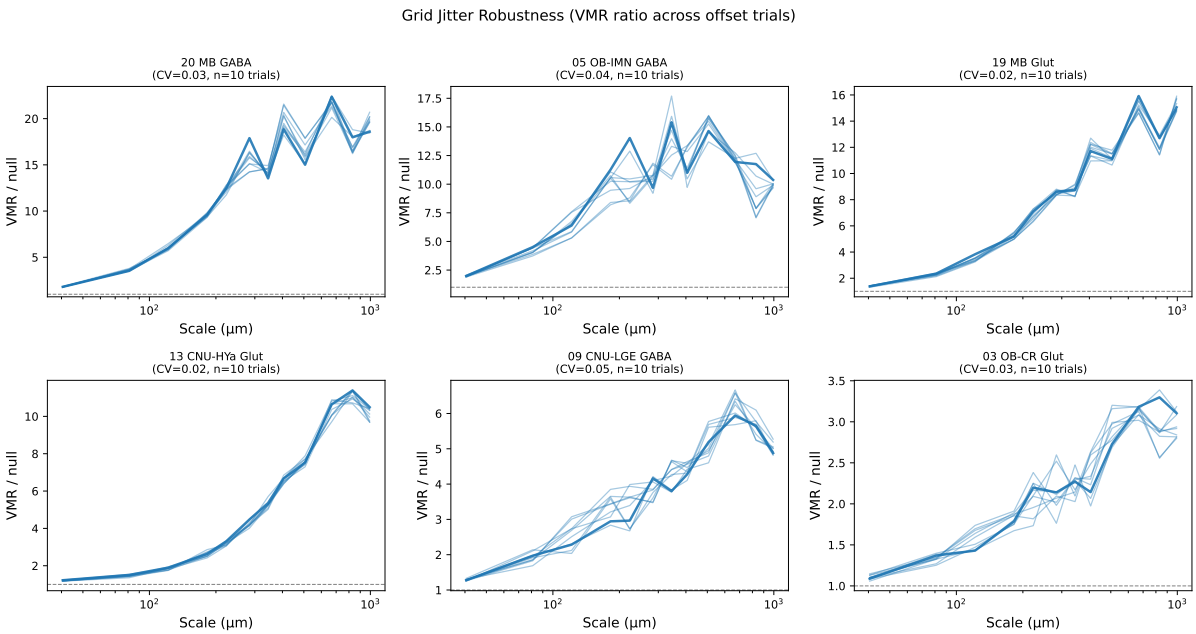

Figure 3: Grid placement robustness. Peak VMR ratios computed across 10 random grid origin offsets show CV $< 5\%$ for all cell types, confirming that fingerprints are independent of arbitrary grid placement.

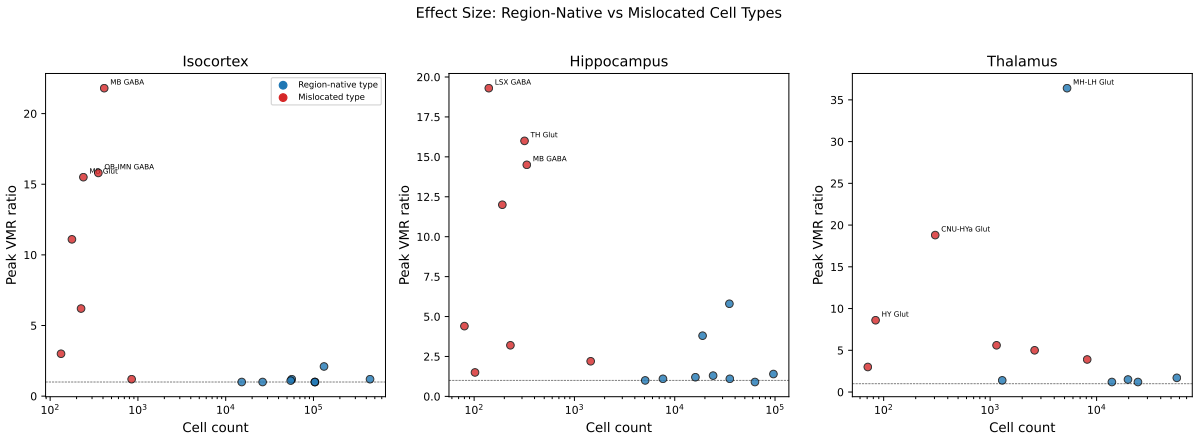

Figure 4: Native versus mislocated cell types. Types classified as "mislocated" (present in a region but primarily assigned to an adjacent region) show dramatically higher effect sizes due to tight spatial confinement at their primary location. Separating these from region-native types reveals the biologically interpretable range of spatial organization.

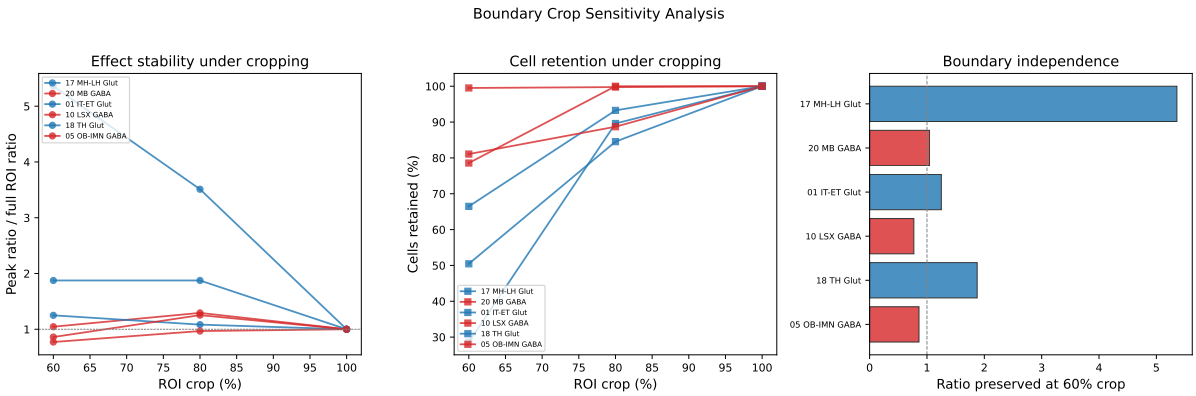

Figure 5: Boundary crop stability. Fingerprints computed on progressively cropped interior regions remain stable, indicating that results are not driven by edge effects or boundary artifacts.

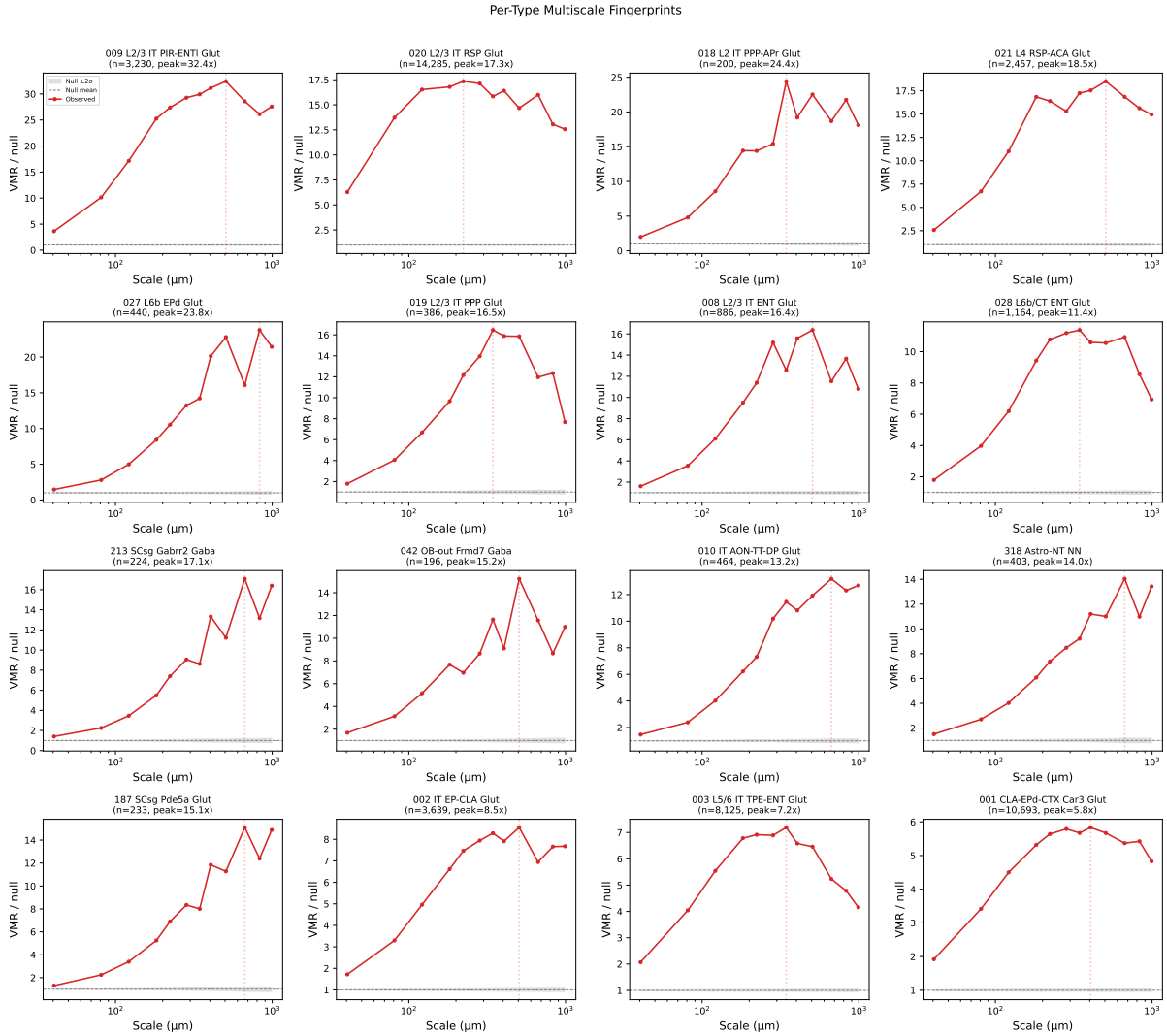

Figure 6: Subclass-level VMR fingerprints in isocortex. Excitatory subtypes (IT, ET, CT, L6b) show layer-specific characteristic clustering scales, resolving the expected laminar hierarchy. Glial and vascular types show small-scale organization consistent with local microenvironment patterning.

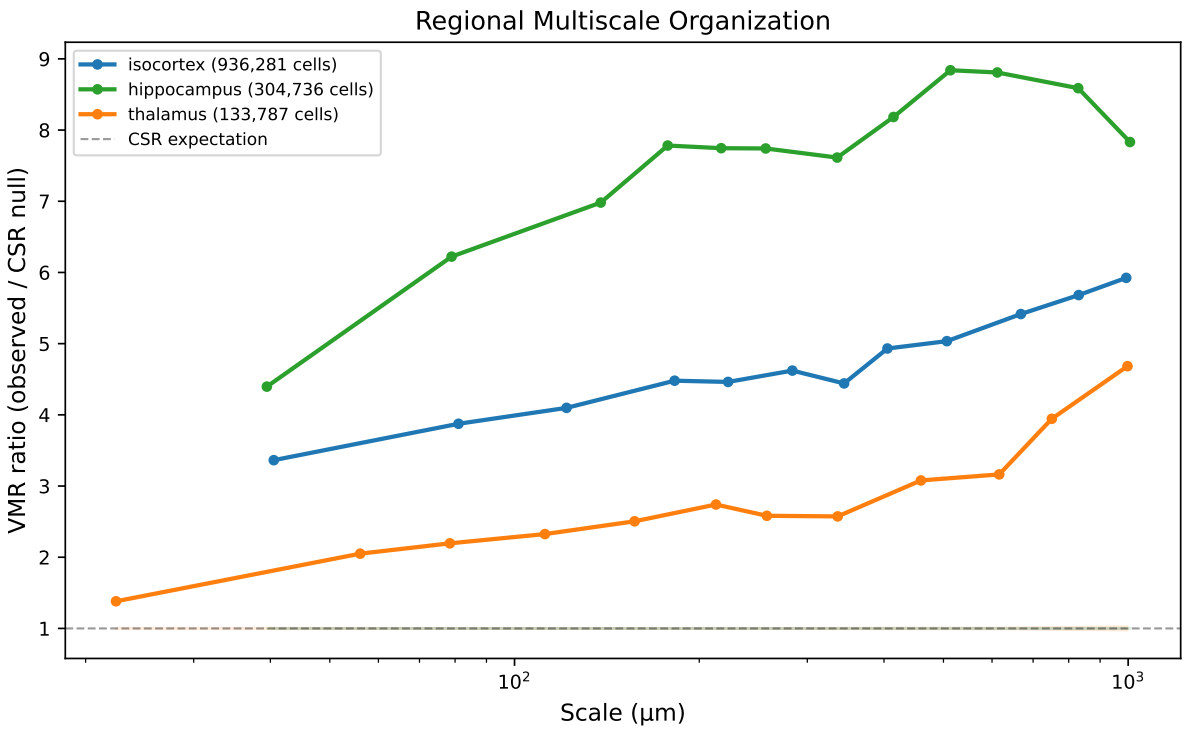

Figure 7: Regional comparison of spatial fingerprints across isocortex, hippocampus, and thalamus. Shared cell types show region-dependent fingerprint variations, reflecting differences in local tissue architecture. Hippocampal fingerprints show stronger mesoscale compartmentalization than isocortex, consistent with dentate gyrus and CA subfield structure.

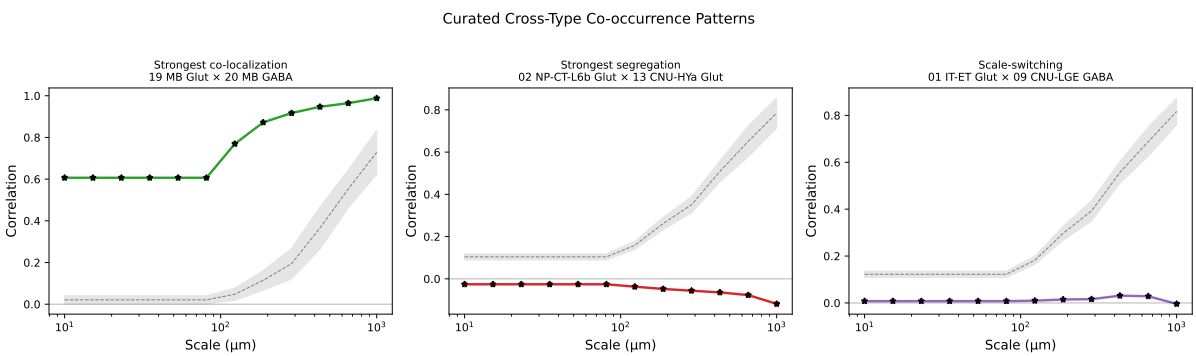

Figure 8: Scale-dependent cross-type co-occurrence in isocortex. Curated pairs of cell types show Pearson correlation of binned counts across spatial scales. Cortical layer types are overwhelmingly co-localized (96–98% of pairs positive at all scales), consistent with laminar stacking within shared cortical columns.

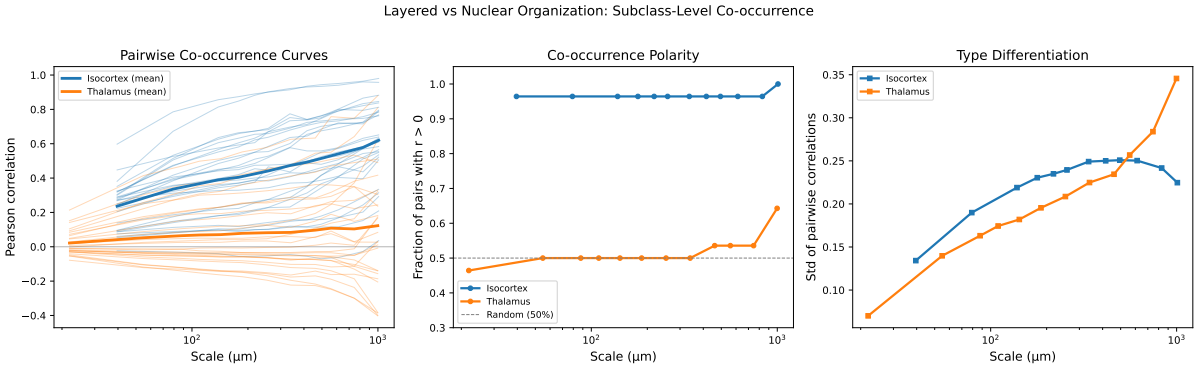

Figure 9: Layered versus nuclear organization. Comparison of cross-type co-occurrence patterns between isocortex (layered architecture) and thalamus (nuclear architecture). Isocortical types show predominantly positive co-localization at all scales, while thalamic types show mixed, weak associations with more frequent scale-switching—consistent with non-overlapping nuclear territories versus shared laminar columns.

with local microenvironment patterning. Hippocampal fingerprints show stronger mesoscale compartmentalization than isocortex, consistent with the highly structured dentate gyrus and CA subfield architecture.

Layered versus nuclear organization. The subclass-level co-occurrence comparison between isocortex and thalamus reveals qualitatively distinct organizational motifs. Cortical layer types are overwhelmingly co-localized at all scales (96–98% of pairs positive), consistent with laminar stacking within shared cortical columns. In contrast, thalamic nuclear types show weak, mixed correlations (49–57% positive), with more frequent scale-switching—consistent with a compartmental architecture where distinct nuclei occupy non-overlapping territories. This suggests that the framework can distinguish between fundamentally different modes of tissue organization.

Universality across biological scales. The same VMR/covariance-aware chi-squared framework has now been applied to three distinct biological systems: protein nanoclusters at nm resolution (SMLM), synaptic input clustering at ţm resolution (dendritic trees), and cell type organization at mm resolution (tissue sections). In each case, the framework detects meaningful, scale-specific spatial structure with appropriate null models. This convergence supports the concept of a universal spatial fingerprinting toolkit for biology.

Limitations. Several limitations should be noted. First, in datasets of this size, nearly all types are statistically significant, making p-values uninformative for distinguishing biologically meaningful from trivial effects; we address this by reporting effect sizes throughout. Second, Pearson correlation for co-occurrence is simple but may miss more complex spatial relationships; future work could incorporate mark correlation functions or cross-K statistics. Third, our analysis uses 2D projections of a 3D brain; extension to full 3D spatial fingerprints would be valuable for datasets with sufficient axial resolution.

Future directions. A systematic subclass-level analysis across all brain regions—beyond the isocortex and thalamus focus here—could test whether the layered/nuclear distinction extends to other organizational motifs (e.g., the nuclear but layered cerebellar cortex). Integration with gene expression profiles could reveal whether spatial fingerprint similarity predicts transcriptomic similarity.

Data Availability

The Allen Brain Cell Atlas MERFISH dataset is publicly available at `https://portal.brain-map.org/atlases-and-data/bkp/abc-atlas`. All code, computed results, and figures are available at `https://github.com/saivoru777-ship-it/spatial-transcriptomics-fingerprints`.

Acknowledgments

We thank the Allen Institute for Brain Science for making the MERFISH dataset publicly available. This work used the `abc_atlas_access` Python package for data retrieval.

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